

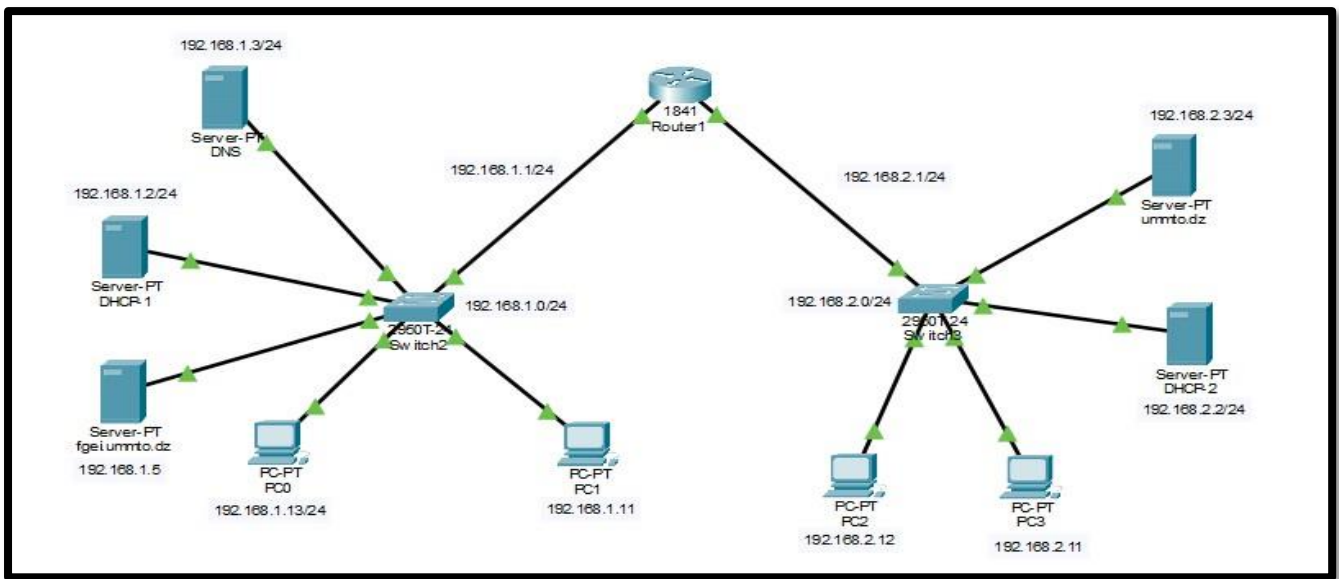
# Lab Report – Router Configuration

## (DHCP + LAN Interconnection)

Last Name : Belkacem

First Name : Nacim

### GRP : 1.2



### Definitions of DNS and DHCP Protocols :

DNS (Domain Name System) is an application layer protocol that converts a domain name into an IP address. When a user enters a URL, the DNS server provides the corresponding IP address to allow access to the website.

DHCP (Dynamic Host Configuration Protocol) is a protocol that automatically assigns network settings (IP address, subnet mask, gateway, and DNS) to devices on a TCP/IP network. It simplifies configuration and reduces errors.

### Objective of the Lab :

The objective of this lab is to understand and configure DNS and DHCP services in a simulated topology using Cisco Packet Tracer, consisting of two DHCP servers, one DNS server, two web servers, four hosts, switches, and a router.

### Configuration Steps:

- **Router Configuration:**

- Interface Fa0/0: IP Address: 192.168.1.1  
Subnet Mask: 255.255.255.0

Interface Fa0/1: IP Address: 192.168.2.1  
Subnet Mask: 255.255.255.0

- Configuration Code:

```
Router>en
```

```
Router#conf t
```

```
Router (config)#int fa0/0
```

```
Router(config-if)#ip address 192.168.1.1 255.255.255.0
```

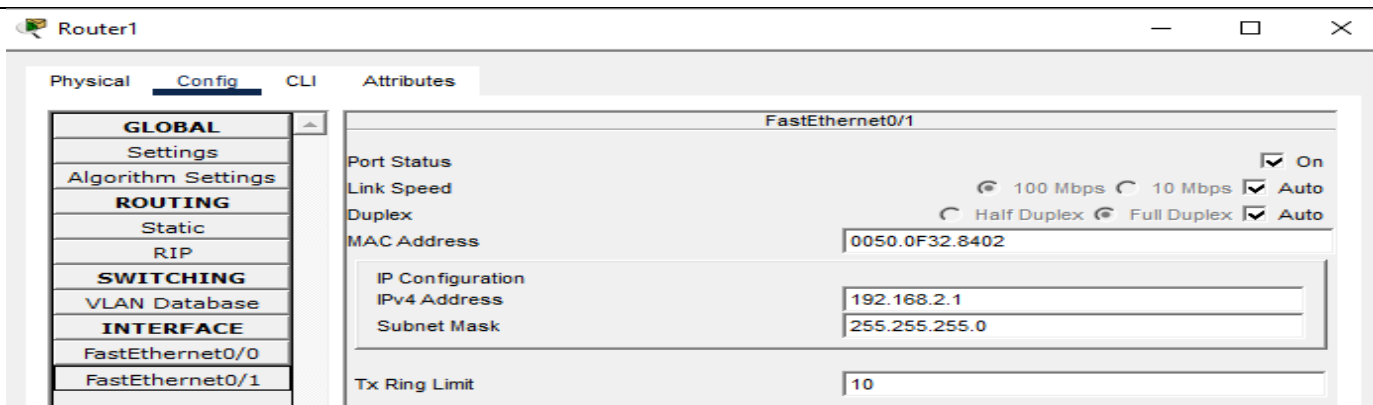
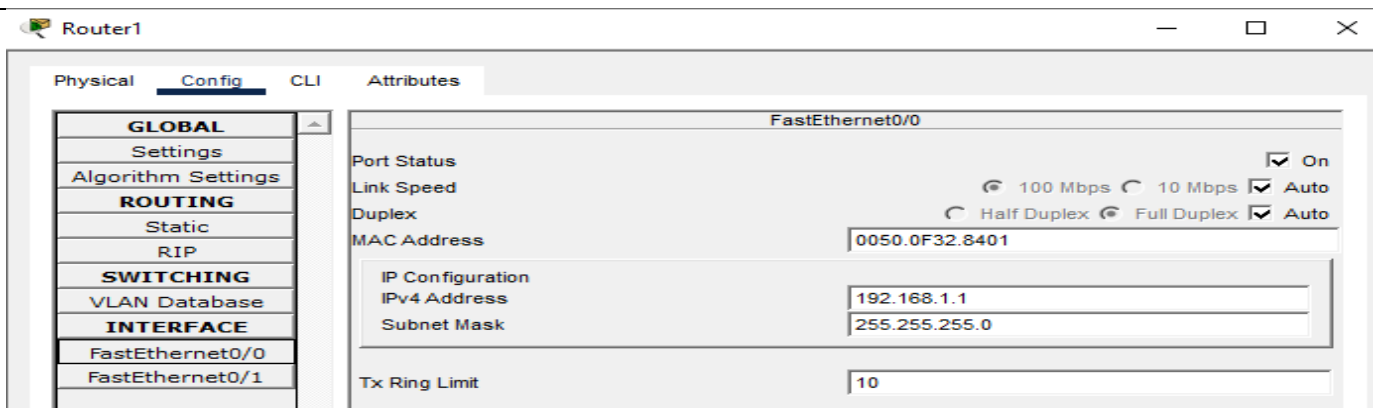
```
Router (config-if)#no sh
```

```
Router(config-if)#exit
```

```
Router (config)#int fa0/1
```

```
Router (config-if)#ip address 192.168.2.1 255.255.255.0
```

```
Router (config-if)#no sh
```



## DHCP Servers Configuration:

- The **DHCP-1** server, belonging to the **192.168.1.0/24** subnet, was configured with the address **192.168.1.2**, a default gateway **192.168.1.1**, and a dynamic range starting from **192.168.1.10**, using the DNS **192.168.1.3**.
- The **DHCP-2** server, for the **192.168.2.0/24** subnet, was assigned the address **192.168.2.2**, a default gateway **192.168.2.1**, and a dynamic range starting from **192.168.2.6**, using the same DNS server.
- All hosts were configured in DHCP mode and automatically received their network settings.

DHCP-1

Physical
Config
Services
Desktop
Programming
Attributes

IP Configuration

☐ DHCP
☒ Static

IPv4 Address
192.168.1.2

Subnet Mask
255.255.255.0

Default Gateway
192.168.1.1

DNS Server
192.168.1.3

DHCP-2

Physical
Config
Services
Desktop
Programming
Attributes

IP Configuration

☐ DHCP
☒ Static

IPv4 Address
192.168.2.2

Subnet Mask
255.255.255.0

Default Gateway
192.168.2.1

DNS Server
192.168.1.3

DHCP-1

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

DHCP

Interface
FastEthernet0
Service
☒ On
☐ Off

Pool Name
serverPool1

Default Gateway
192.168.1.1

DNS Server
192.168.1.3

Start IP Address :
192
168
1
10

Subnet Mask:
255
255
255
0

Maximum Number of Users :
50

TFTP Server:
0.0.0.0

WLC Address:
0.0.0.0

Add
Save
Remove

| Pool Name   | Default Gateway | DNS Server  | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|-------------|-----------------|-------------|------------------|-------------|----------|-------------|-------------|
| serverPool1 | 192.168....     | 192.168.... | 192.168....      | 255.255.... | 50       | 0.0.0.0     | 0.0.0.0     |
| serverPool  | 0.0.0.0         | 0.0.0.0     | 192.168....      | 255.255.... | 512      | 0.0.0.0     | 0.0.0.0     |

DHCP-2

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

DHCP

Interface
FastEthernet0
Service
☒ On
☐ Off

Pool Name
serverPool2

Default Gateway
192.168.2.1

DNS Server
192.168.1.3

Start IP Address :
192
168
2
10

Subnet Mask:
255
255
255
0

Maximum Number of Users :
50

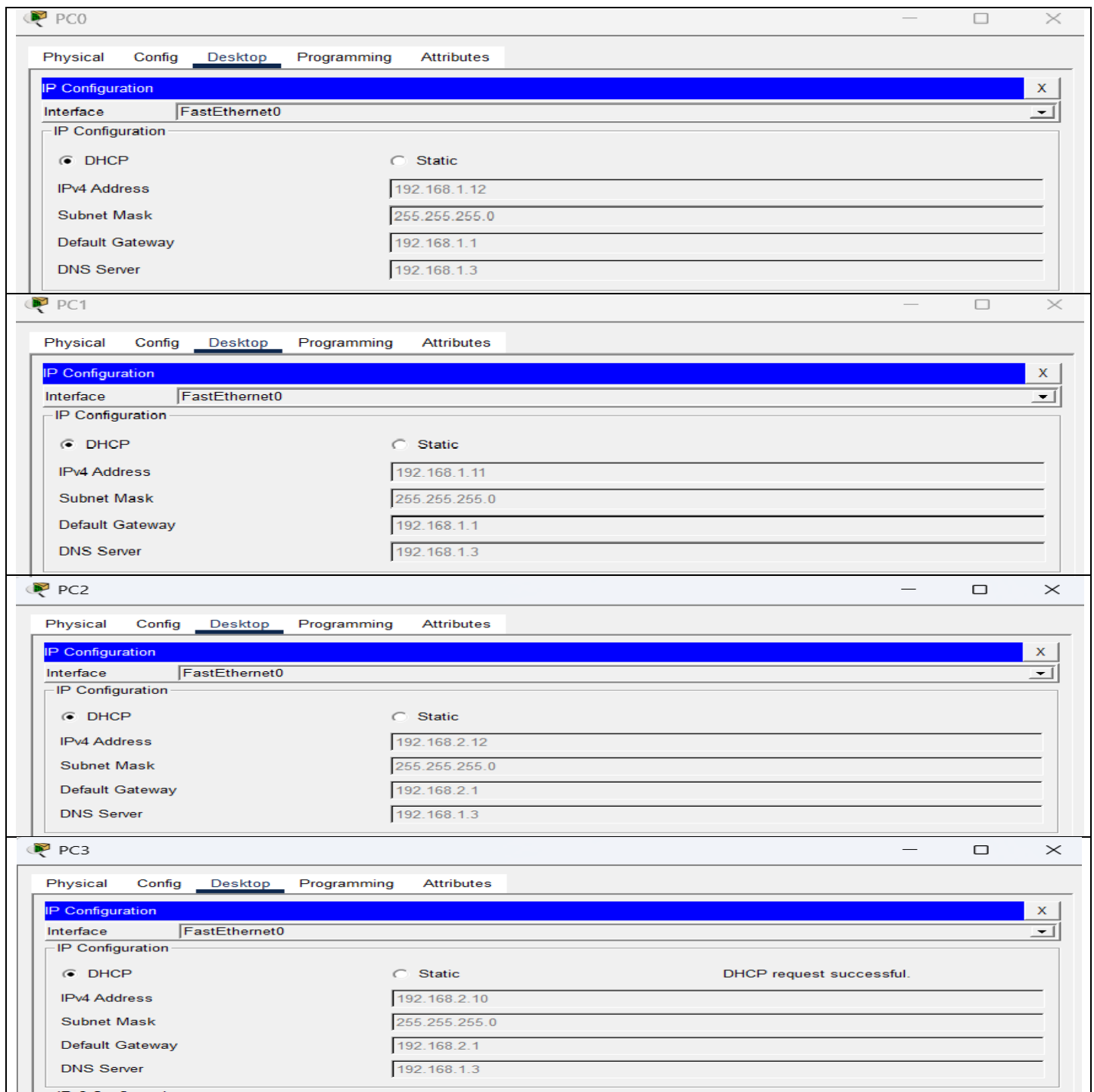
TFTP Server:
0.0.0.0

WLC Address:
0.0.0.0

Add
Save
Remove

| Pool Name   | Default Gateway | DNS Server  | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|-------------|-----------------|-------------|------------------|-------------|----------|-------------|-------------|
| serverPool2 | 192.168....     | 192.168.... | 192.168....      | 255.255.... | 50       | 0.0.0.0     | 0.0.0.0     |
| serverPool  | 0.0.0.0         | 0.0.0.0     | 192.168....      | 255.255.... | 512      | 0.0.0.0     | 0.0.0.0     |

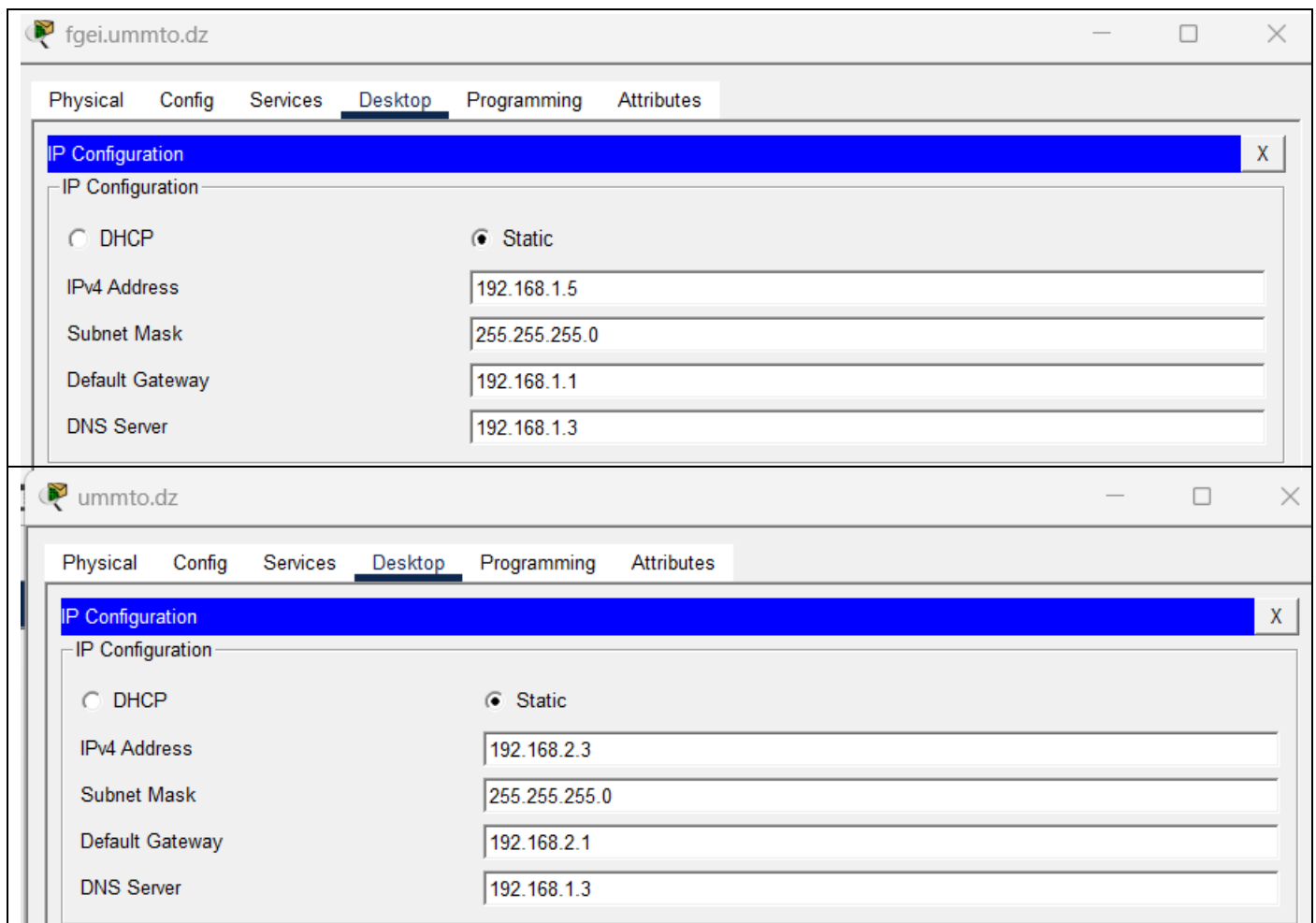
## Enable DHCP on all hosts :



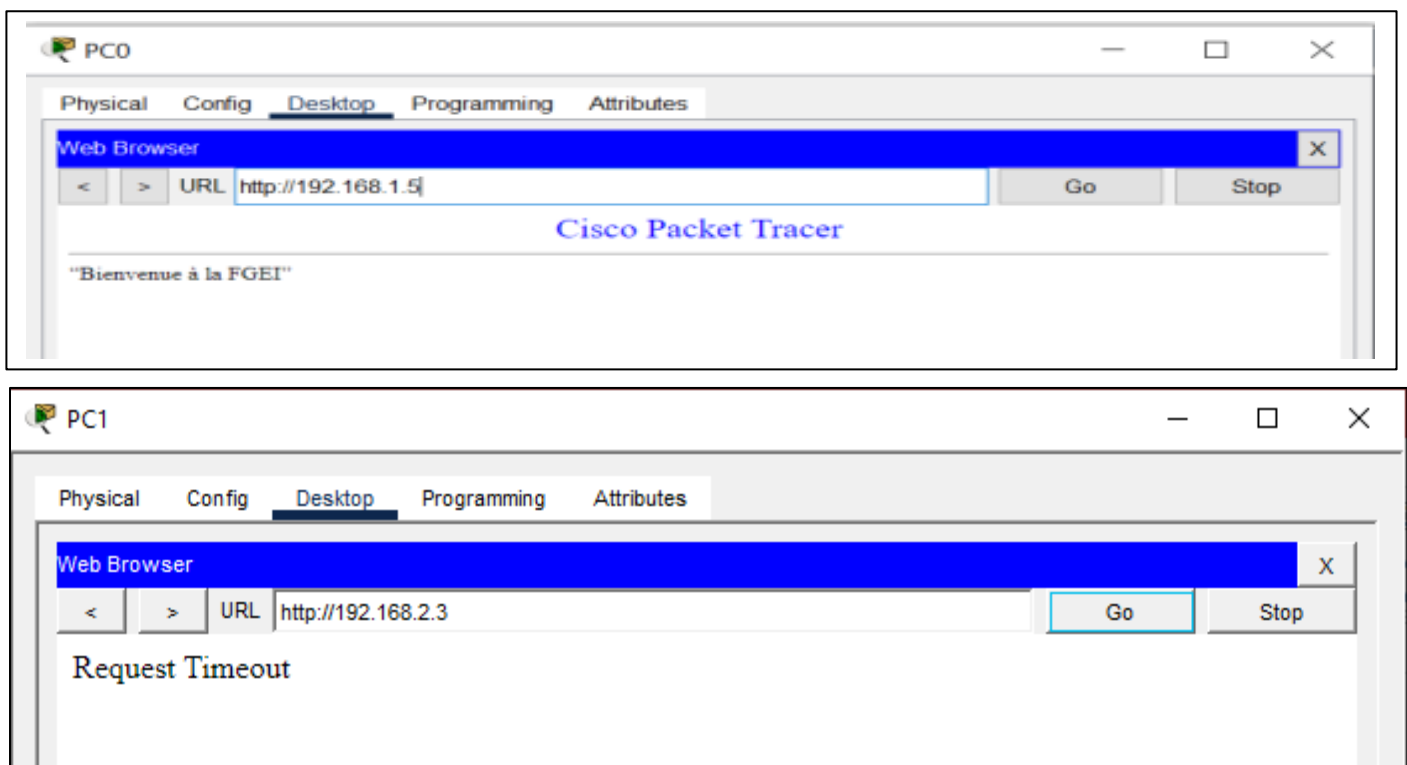
## Web Servers Configuration:

Two web servers were configured in the topology, each with a static IP address and a customized HTML page.

- **Web 1 (192.168.1.5)**, corresponding to the domain **fgei.ummtto.dz**, shows **"Request Timeout"** because the router has not yet been configured to connect the two web pages.
- **Web 2 (192.168.2.3)**, associated with the domain **ummtto.dz**, displays the message **"Bienvenue à la FGEI"**.



### Test IP Address in the host :



## DNS Server Configuration:

### **Activate the DNS Server :**

- Configure the DNS server's IP address and default gateway.
  - IP Address: 192.168.1.3
  - Default Gateway: 192.168.1.1
  - Subnet Mask: 255.255.255.0
- Activate the DNS service on the server.

### **2. Add a DNS Record :**

- Domain Name: www.ummto.dz and fgei.ummto.dz
- IP Address: that of the corresponding web server

### **3. Update DHCP Servers :**

- Add the DNS server's IP address to the settings of DHCP-1 and DHCP-2 so that clients use it automatically.

The screenshot shows the 'DNS1' configuration window with the 'Services' tab selected. The 'DNS' service is turned 'On'. Under 'Resource Records', two A records are listed:

| No. | Name          | Type     | Detail      |
|-----|---------------|----------|-------------|
| 0   | fgei.ummto.dz | A Record | 192.168.1.5 |
| 1   | www.ummto.dz  | A Record | 192.168.2.3 |

## **Activate HTTP on the two web servers:**

The screenshot shows the 'fgei.ummto.dz' configuration window with the 'Services' tab selected. The 'HTTP' service is turned 'On'. Below the service controls is a 'File Manager' table:

|   | File Name             | Edit   | Delete   |
|---|-----------------------|--------|----------|
| 1 | copyrights.html       | (edit) | (delete) |
| 2 | cscoptlogo177x111.jpg |        | (delete) |
| 3 | helloworld.html       | (edit) | (delete) |
| 4 | image.html            | (edit) | (delete) |
| 5 | index.html            | (edit) | (delete) |

The screenshot shows the 'ummto.dz' configuration window with the 'Services' tab selected. The 'HTTP' service is turned 'On'. Below the service controls is a 'File Manager' table:

|   | File Name             | Edit   | Delete   |
|---|-----------------------|--------|----------|
| 1 | copyrights.html       | (edit) | (delete) |
| 2 | cscoptlogo177x111.jpg |        | (delete) |
| 3 | helloworld.html       | (edit) | (delete) |
| 4 | image.html            | (edit) | (delete) |
| 5 | index.html            | (edit) | (delete) |

## Final Router Configuration to Connect Switches, Web Servers, and Hosts Together:

```
Router> enable
Router# configure terminal
Router(config)# int fa0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# int fa0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10
Router(config)# ip dhcp pool serverPool1
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 192.168.1.3
Router(dhcp-config)# exit
Router(config)# ip dhcp pool serverPool2
Router(dhcp-config)# network 192.168.2.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.2.1
Router(dhcp-config)# dns-server 192.168.1.3
Router(dhcp-config)# exit
Router(config)# end
```

## Test Ping All Host :

| Fire                                                                                | Last Status | Source | Destination | Type | Color                                                                               | Time(sec) | Periodic | Num | Edit   | Delete |
|-------------------------------------------------------------------------------------|-------------|--------|-------------|------|-------------------------------------------------------------------------------------|-----------|----------|-----|--------|--------|
|  | Successful  | PC0    | PC2         | ICMP |  | 0.000     | N        | 0   | (edit) |        |
|  | Successful  | PC0    | PC3         | ICMP |  | 0.000     | N        | 1   | (edit) |        |
|  | Successful  | PC1    | PC2         | ICMP |  | 0.000     | N        | 2   | (edit) |        |
|  | Successful  | PC1    | PC3         | ICMP |  | 0.000     | N        | 3   | (edit) |        |

## DNS Resolution Test:

- From the clients' web browsers, enter the addresses: `www.ummto.dz` OR `fgei.ummto.dz`.
- The DNS server returns the IP address of the corresponding web server. The page displays "`ummto.dz`" or "`fgei.ummto.dz`", thanks to the HTML code added to the web servers, confirming that the DNS resolution works correctly.

### Access from PC-0 Connected to Switch 1 to the Web Interface on Switch 2 :



### Access from PC-03 Connected to Switch 2 to the Web Interface on Switch 1 :



## **Conclusion:**

This lab highlighted the key role of DNS and DHCP in managing a network. DHCP automatically assigns IP addresses, subnet masks, and gateways, simplifying network administration. DNS translates domain names into IP addresses, making web access easier.

The setup of routers, DHCP and DNS servers, and web servers, along with the connectivity and DNS resolution tests, confirmed that the network operates correctly and efficiently.